Complete local counts at prescribed initial positions in Erdős–Straus decompositions

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Abstract

For primes $p \equiv 1 \pmod 4$, we count all unordered Type I and Type II Erdős–Straus decompositions at prescribed initial smallest denominators. Put $\delta = 2^{-20}$ and $\kappa = (\log 3)/8 - \delta$. Outside one set of $O(X/(\log X)^{1+\delta})$ primes up to X, uniformly for $a \equiv 3 \pmod 4$, $3 \le a \le (\log p)^{\delta}$, and $x = (p+a)/4$, the two counts are

$$F_{\mathrm{I}}(p,x) = \frac{\tau(x^2)}{\varphi(a)}\big(1 + O((\log p)^{-\kappa})\big), \qquad F_{\mathrm{II}}(p,x) = \frac{\tau(x^2)}{2\varphi(a)}\big(1 + O((\log p)^{-\kappa})\big).$$

On a possibly smaller common good set with the same exceptional-set order, each count is bounded below by a constant multiple of $(\log p)^{\theta \log 3}/\varphi(a)$ for any fixed $0 < \theta < 15/16$. The argument combines established divisor parametrizations, finite-group character estimates, and uniform distribution estimates for shifted primes. We give both a direct support-compressed moment proof and a uniform application of Elliott's Bernoulli model.

Keywords. Egyptian fractions; prescribed denominators; local multiplicities; shifted primes; divisor distributions; Bombieri–Vinogradov theorem.

1 Introduction and main results

The Erdős–Straus equation is

$$\frac{4}{p} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}, \qquad x, y, z \in \mathbb{Z}_{>0}. \tag{1}$$

We take $p \equiv 1 \pmod 4$ prime and prescribe initial positions $x = (p+a)/4$ with $a = 3 + 4j$. Solutions are counted without regard to the order of their denominators. Write $F_{\mathrm{I}}(p,x)$, respectively $F_{\mathrm{II}}(p,x)$, for the number of solutions with strictly smallest denominator x and precisely one, respectively two, denominators divisible by p. All logarithms are natural, and $\tau(n)$ denotes the number of positive divisors of n.

For $3 \le a < p$, $a \equiv 3 \pmod 4$, we have

$$(a,p) = (a,x) = 1, \qquad x < p/2, \qquad p \nmid x.$$

Reduction of the cleared equation modulo p shows that any solution at this x has at least one of its other denominators divisible by p. Thus the two types exhaust the complete local count at this position.

Theorem 1.1 (Complete local counts)**.** *Set*

$$\delta = 2^{-20}, \qquad \kappa = \frac{\log 3}{8} - \delta > 0,$$
$$J(p) = \left\lfloor \frac{(\log p)^{\delta} + 1}{4} \right\rfloor, \qquad a_j = 3 + 4j, \quad x_j = (p + a_j)/4.$$

There is a set $\mathcal{E}$ of primes $p \equiv 1 \pmod 4$ such that

$$\#\{p \le X : p \in \mathcal{E}\} \ll \frac{X}{(\log X)^{1+\delta}} \qquad (X \to \infty), \tag{2}$$

and, for sufficiently large $p \notin \mathcal{E}$, simultaneously at all positions $0 \le j < J(p)$,

$$F_{\mathrm{I}}(p, x_j) = \frac{\tau(x_j^2)}{\varphi(a_j)} \left(1 + O((\log p)^{-\kappa})\right), \tag{3}$$

$$F_{\mathrm{II}}(p, x_j) = \frac{\tau(x_j^2)}{2\varphi(a_j)} \left(1 + O((\log p)^{-\kappa})\right). \tag{4}$$

The error constants are uniform in j. In particular,

$$\begin{aligned} \frac{F_{\mathrm{I}}(p, x_j)}{F_{\mathrm{II}}(p, x_j)} &= 2 + O((\log p)^{-\kappa}), \\ F_{\mathrm{I}}(p, x_j) + F_{\mathrm{II}}(p, x_j) &= \frac{3\tau(x_j^2)}{2\varphi(a_j)} \left(1 + O((\log p)^{-\kappa})\right). \end{aligned} \tag{5}$$

Solutions at different positions j are distinct.

Corollary 1.2 (Quantitative multiplicity)**.** *Fix $0 < \theta < 15/16$ and put $\gamma_\theta = \theta \log 3$. There is an exceptional set $\mathcal{E}_\theta \supseteq \mathcal{E}$ with*

$$\#\{p \le X : p \in \mathcal{E}_\theta\} \ll_\theta X/(\log X)^{1+\delta}$$

such that, for sufficiently large $p \notin \mathcal{E}_\theta$, all positions $0 \le j < J(p)$ satisfy Theorem 1.1 and

$$F_{\mathrm{I}}(p, x_j) \ge \frac{(\log p)^{\gamma_\theta}}{4\varphi(a_j)}, \qquad F_{\mathrm{II}}(p, x_j) \ge \frac{(\log p)^{\gamma_\theta}}{8\varphi(a_j)}. \tag{6}$$

In particular, one may take $\gamma_* = \gamma_{7/8} = 7 \log 3/8 = 0.96128575\ldots$.

The enlarged exceptional set comes from one additional total-factor statistic per shift. Its order is unchanged, but it need not be the same set point by point. The range $\theta < 15/16$ is sufficient for the chosen proof parameters; no endpoint or optimality assertion is made.

Corollary 1.3 (Fixed initial catalogue)**.** *For each fixed integer $J_0 \ge 1$, the local asymptotics (3)–(4) hold simultaneously for $0 \le j < J_0$ outside an exceptional set of $O_{J_0}(X/(\log X)^{1+\delta})$ primes $p \le X$, $p \equiv 1 \pmod 4$. For each additional fixed $0 < \theta < 15/16$, the bounds (6) hold for these positions outside a possibly enlarged exceptional set of size $O_{J_0,\theta}(X/(\log X)^{1+\delta})$.*

The constants are deliberately unoptimized. The catalogue becomes nonempty precisely when

$$J(p) \ge 1 \quad \Longleftrightarrow \quad p \ge \exp(3^{2^{20}}). \tag{7}$$

The results are unconditional and asymptotic, with a further ineffective onset inherited from Siegel–Walfisz. The exceptional set is defined by sufficient factor-distribution tests. The theorem leaves the individual exceptional primes untreated and gives no complexity bound for constructing the decompositions.

1.1 Earlier results and the quantitative application

Bradford's Propositions 1–4 [4] give complete fixed-denominator divisor criteria for both types. We use these criteria in a normalized form and include their inverse maps to fix the counting conventions. Elsholtz and Tao [6, Theorem 1.8 and Section 6] already obtain many solutions at the first position $x = (p+3)/4$ for a relative-density-one set of primes. Their fixed-shift additive-function theorem [6, Appendix, Lemma A.3], together with the complete divisor criteria and character orthogonality, also gives complete local asymptotics on any fixed finite catalogue. It yields multiplicity exponents arbitrarily close to $\log 3$ on such a catalogue. Appendix B records this deduction explicitly.

Hall's work on the divisors of $p-1$ [8] and the per-class divisor estimates in the proof of Bettin et al., Lemma 4.9 [7], provide further fixed-shift precedents. The latter also give a one-sided Type II lower bound of exponent $\log 2 - \varepsilon$ on every fixed finite catalogue; the substitution is given in Appendix B. A diagonal choice from fixed-catalogue results allows some unspecified slowly growing catalogue. Thus the quantitative content considered here is the prescribed logarithmic-power length, a uniform relative error for both complete counts, and one common exceptional-set estimate.

Other nearby results concern different counting questions. Elsholtz and Planitzer [2] obtain lower bounds for total solution counts on density-one sets of integers and on infinite sets of primes. Pomerance and Weingartner [11] bound exceptional sets for existence of a decomposition with a varying numerator. Dahan [5, Section 4.3] allows a growing search depth in a Type II witness family. Complete parametrizations are also studied in Bello–Hernández, Benito, and Fernández [12], while Chamberland [3] gives a Type II prime-representation criterion. These conclusions concern total counts, existence, or search depth; the present theorems count each prescribed position.

Our proof uses standard distribution results for primes. A direct moment expansion groups terms by distinct squarefree supports. The normalized coefficient of each support is polylogarithmic, and its progression modulus is at most $4N^{1/4}$, so ordinary Bombieri–Vinogradov controls the accumulated error. Alternatively, Elliott's established Bernoulli model, in the form recalled by Gorodetsky [10, Theorem 1.1], gives the same transfer after a uniform remainder calculation; Appendix C supplies that calculation. Ford's Poisson model for fixed shifts [9] is another relevant distributional framework. The application here combines these analytic inputs with complete prime-power divisor boxes and the exact Type I/Type II multiplicities.

2 Complete local counts and character mixing

2.1 The square-divisor dictionary

For this subsection, assume $p \equiv 1 \pmod 4$ is prime, $a \equiv 3 \pmod 4$, $3 \le a < p$, and $x = (p+a)/4$. In particular $(a,x) = (a,p) = 1$ and $x < p/2$.

For $b \in (\mathbb{Z}/a\mathbb{Z})^\times$, define

$$C_{a,x}(b) = \#\{d \mid x^2 : d\,x^{-1} \equiv b \pmod a\}, \qquad M(x) = \tau(x^2). \tag{8}$$

The inverse in (8) is taken modulo a.

Lemma 2.1 (Divisors and primitive ratios)**.** *The divisors $d \mid x^2$ correspond bijectively to ordered pairs (u,v) with $(u,v) = 1$ and $uv \mid x$. The correspondence is*

$$g = (d,x), \quad u = d/g, \quad v = x/g, \quad D = g^2/d, \qquad x = Duv, \quad d = Du^2.$$

In particular, D is a positive integer and $d/x = u/v$. If $x = \prod_{q\mid x} q^{e_q}$ and $d = \prod_{q\mid x} q^{j_q}$, the ratio exponent at q is $j_q - e_q \in \{-e_q, \dots, e_q\}$.

Proof. At each prime, the exponents of u, v, D are respectively $\max(j_q - e_q, 0)$, $\max(e_q - j_q, 0)$, and $2\min(j_q, e_q) - j_q$. The last is nonnegative because $0 \le j_q \le 2e_q$. These formulas give all the claimed identities. Conversely, a coprime pair with $uv \mid x$ determines $D = x/(uv)$ and $d = Du^2$, which divides x^2 and reverses the construction. □

Proposition 2.2 (Exact counts)**.** *In the stated range,*

$$F_{\mathrm{I}}(p, x) = C_{a,x}(-p^{-1}), \qquad F_{\mathrm{II}}(p, x) = \frac{1}{2} C_{a,x}(-1). \tag{9}$$

Both identities count all solutions with strictly smallest denominator x.

Proof. For Type I, the condition $dx^{-1} \equiv -p^{-1} \pmod a$ is equivalent to $pd + x \equiv 0 \pmod a$, or to $4d + 1 \equiv 0 \pmod a$. Since $dd' = x^2$ for $d' = x^2/d$, it also implies $d' + px \equiv 0 \pmod a$. Set

$$y = \frac{px + x^2/d}{a}, \qquad z = \frac{p(x + pd)}{a}. \tag{10}$$

Equivalently, with Lemma 2.1,

$$c_1 = \frac{pu + v}{a}, \qquad (x, y, z) = (Duv, Dvc_1, pDuc_1). \tag{11}$$

Indeed,

$$\frac{1}{y} + \frac{1}{z} = \frac{pu + v}{pDuvc_1} = \frac{a}{px}, \qquad \frac{1}{x} + \frac{a}{px} = \frac{4}{p}.$$

Moreover $p \nmid Duv$ and $ac_1 \equiv v \not\equiv 0 \pmod p$, so only z is divisible by p. We have

$$\frac{y}{x} = \frac{pu + v}{au} > \frac{p}{a} > 1, \qquad \frac{z}{y} = \frac{pu}{v} \ge \frac{p}{x} > 2.$$

Thus $x < y < z$. The ordered pair (u, v) is recovered as the reduced ratio $z/(py)$, or d is recovered from $d = (az/p - x)/p$. There is no division by two.

For completeness, take any Type I solution and name the non-p second denominator y and the p-divisible one pV. The reciprocal identity is $ayV = pxV + xy$. It implies $aV - x > 0$ and, after reduction modulo p, $p \mid aV - x$. Write $aV - x = pd$ with $d > 0$. Substitution gives

$$(ay - px)d = x^2.$$

Hence $d \mid x^2$, $pd + x \equiv 0 \pmod a$, and the solution is (10). The inequalities already proved also show that its p-divisible denominator is the larger one.

For Type II write the two other denominators as pY, pZ. Then

$$\frac{a}{x} = \frac{1}{Y} + \frac{1}{Z}, \qquad (aY - x)(aZ - x) = x^2.$$

Both factors $aY - x$ and $aZ - x$ are positive by the reciprocal identity. This gives all divisors $d \mid x^2$ with $d + x \equiv 0 \pmod a$, by

$$\{y, z\} = \left\{ \frac{p(x + d)}{a}, \frac{p(x + x^2/d)}{a} \right\}. \tag{12}$$

The two denominators are positive multiples of p, so they exceed $x < p/2$. Swapping them sends d to x^2/d. A fixed point would be $d = x$, which would require $a \mid 2x$, impossible because $(a, x) = 1$ and $a \ge 3$ is odd. The fibers therefore have size exactly two. In primitive variables this is the familiar recovery $h = (u + v)/a$ and $\{y, z\} = \{pDvh, pDuh\}$. □

These criteria are equivalent to established fixed-x divisor parametrizations; compare Bradford [4, Propositions 1–4]. For his Type I convention the divisor is x^2/d in our notation.

2.2 Mixing in the complete prime-power box

Lemma 2.3 (Full divisor-box mixing)**.** *Let a, x be positive integers with $(a, x) = 1$, and let $m \geq 0$. Suppose that for every nonprincipal character $\chi \bmod a$, at least m distinct prime divisors $q \mid x$ satisfy $\Re\chi(q) \leq 0$. Then, for every unit class $b \bmod a$,*

$$\left|\frac{\varphi(a)C_{a,x}(b)}{\tau(x^2)} - 1\right| \leq (\varphi(a) - 1)3^{-m}. \tag{13}$$

The counted prime divisors may be drawn from any chosen subwindow. All additional primes and prime powers are included in $C_{a,x}$.

Proof. For $|z| = 1$ and $e \geq 1$ let

$$A_e(z) = \frac{1}{2e+1}\sum_{j=-e}^{e} z^j.$$

Always $|A_e(z)| \leq 1$. If $\Re z \leq 0$, then for $e = 1$, $|A_1(z)| = |1 + 2\Re z|/3 \leq 1/3$. For $e \geq 2$, write $z = e^{it}$. Since $|\sin(t/2)| \geq 1/\sqrt{2}$, the finite geometric sum gives

$$|A_e(z)| = \frac{|\sin((2e+1)t/2)|}{(2e+1)|\sin(t/2)|} \leq \frac{\sqrt{2}}{2e+1} \leq \frac{\sqrt{2}}{5} < \frac{1}{3}.$$

Using $x = \prod q^{e_q}$, character orthogonality and the ratio exponents from Lemma 2.1 give

$$\frac{\varphi(a)C_{a,x}(b)}{\tau(x^2)} = \sum_{\chi \bmod a} \overline{\chi(b)} \prod_{q|x} A_{e_q}(\chi(q)).$$

The principal term is 1. Every nonprincipal term has absolute value at most 3^{-m}, by the hypothesis and the preceding factor bounds. Summing proves (13). □

Remark 2.4. A prime power q^e contributes $2e + 1$ possible ratio exponents. The normalized block A_e accounts for all of them; replacing it by e independent ternary choices would instead count 3^e assignments.

3 A common good set for the growing catalogue

3.1 Parameters and sector tests

For sufficiently large N, put

$$\begin{gathered} L = \log\log N, \qquad c = 2^{-14}, \qquad r = \lfloor cL \rfloor, \qquad k = 2r, \\ Q = \exp((\log N)^{1/4}), \qquad Z = N^{1/(8r)}, \\ \mathcal{P}_N = \{N < p \leq 2N : p \text{ prime}, \ p \equiv 1 \pmod 4\}, \\ \mathcal{A}_N = \{3 \leq a \leq 2(\log N)^{\delta} : a \equiv 3 \pmod 4\}. \end{gathered} \tag{14}$$

For $p \in \mathcal{P}_N$, $a \in \mathcal{A}_N$, and nonprincipal $\chi \bmod a$, let

$$x_a = \frac{p+a}{4}, \quad \mathcal{Q}_a(p) = \{Q < q \leq Z : q \text{ prime}, \ q \mid x_a\}, \quad K_a(p) = |\mathcal{Q}_a(p)|,$$

$$Z_{a,\chi}(p) = \#\{q \in \mathcal{Q}_a(p) : \operatorname{Re}\chi(q) \leq 0\}.$$

Define the sufficient-test bad set

$$\mathcal{B}_N = \bigcup_{a \in \mathcal{A}_N} \{p \in \mathcal{P}_N : K_a(p) < L/2\} \ \cup \bigcup_{\substack{a \in \mathcal{A}_N \\ \chi \neq \chi_0 \ (a)}} \{p \in \mathcal{P}_N : Z_{a,\chi}(p) < L/8\}. \tag{15}$$

The complement of $\mathcal{B}_N$ will be the common good set.

Lemma 3.1 (Size of a character sector)**.** *Let G be a finite abelian group and χ a nonprincipal character. Then*

$$S_\chi = \{g \in G : \Re\chi(g) \le 0\}, \qquad |S_\chi| \ge |G|/3.$$

Proof. The image of χ consists of the dth roots of unity for some $d \ge 2$, each with equally many preimages. The number of roots in the closed left half-plane is

$$\lfloor 3d/4 \rfloor - \lceil d/4 \rceil + 1.$$

For $d \ge 6$ this is at least $d/2 - 1 \ge d/3$; for $d = 2, 3, 4, 5$ the counts are $1, 2, 3, 2$, respectively. □

3.2 Distribution inputs and exact comparison means

In the arithmetic sums in this section and their subsequent moment application, the index q always ranges over primes. We record the precise standard distribution inputs. The Siegel–Walfisz theorem implies, for every fixed $H > 0$,

$$\pi(t; a, b) = \frac{\mathrm{Li}(t)}{\varphi(a)} + O_H\left(\frac{t}{(\log t)^H}\right) \quad ((a, b) = 1,\ a \le \log t). \tag{16}$$

Here a fixed Siegel–Walfisz modulus-range parameter $B = 2$ suffices in partial summation; its exponential error absorbs every fixed power $(\log t)^{-H}$. The implied constant and onset may depend on H. We also use the following consequence of the ordinary Bombieri–Vinogradov theorem: for every fixed $A > 0$,

$$\sum_{m \le 4N^{1/4}} \max_{(v,m)=1} \left|\pi(2N; m, v) - \pi(N; m, v) - \frac{\mathrm{Li}(2N) - \mathrm{Li}(N)}{\varphi(m)}\right| \ll_A \frac{N}{(\log N)^A}. \tag{17}$$

These forms follow from the character version of Siegel–Walfisz on p. 2 and the two-term Bombieri–Vinogradov bound on p. 3 of Vaughan's account [13]. Appendix D gives the precise partial-summation steps and parameter substitutions.

For sufficiently large N, the parameters satisfy

$$2(\log N)^\delta < Q < Z < N < p \quad (p \in \mathcal{P}_N). \tag{18}$$

Indeed, $Q < Z$ reduces to $8r < (\log N)^{3/4}$. Also $(a, x_a) = 1$, because a common divisor would divide p, while $a < p$. All window primes are thus units modulo a and exceed a.

For each unit class $b \pmod a$, put

$$\lambda_b = \sum_{\substack{Q < q \le Z \\ q \equiv b \ (a)}} \frac{1}{q - 1}.$$

Partial summation in (16), with $H = 9$, gives uniformly in $a \in \mathcal{A}_N$ and b,

$$\lambda_b = \frac{\log\log Z - \log\log Q}{\varphi(a)} + O((\log Q)^{-8}) + O(Q^{-1}). \tag{19}$$

To justify the uniformity, $a \le \log t$ holds eventually for all $t \ge Q$. Replacing $1/q$ by $1/(q-1)$ costs $O(Q^{-1})$. The boundary and integral errors in partial summation are bounded by a constant times $(\log Q)^{-8}$. For a fixed (a, χ), the sector mean sums λ_b over $b \in S_\chi$; the total mean sums over all unit classes. Thus their errors are at most

$$O\Big(\varphi(a)((\log Q)^{-8} + Q^{-1})\Big) = O\Big((\log N)^{\delta-2} + (\log N)^\delta/Q\Big) = o(1),$$

uniformly in a, χ. This is a sum within one statistic, distinct from the later union over different moduli and characters.

For a nonprincipal character $\chi \pmod a$, define actual statistics

$$I_q(p) = \mathbf{1}_{q \mid p+a}, \qquad Z_{a,\chi}(p) = \sum_{\substack{Q<q\le Z \\ q \bmod a \in S_\chi}} I_q(p), \qquad K_a(p) = \sum_{Q<q\le Z} I_q(p). \tag{20}$$

The congruence $q \mid p + a$ is equivalent to $q \mid x_a$. The exact comparison means are sums of $w_q = 1/(q-1)$ over the respective sets of primes, denoted $\mu_{a,\chi}$ and $\mu_{a,\mathrm{all}}$. They are never replaced by asymptotic expressions inside a high moment.

The difference of logarithms in (19) is

$$\log\log Z - \log\log Q = \frac{3}{4}L - \log(8r). \tag{21}$$

The second term is not bounded as N grows, but it is $o(L)$. Writing $\sigma_\chi = |S_\chi|/\varphi(a) \ge 1/3$, we obtain

$$\mu_{a,\chi} = \sigma_\chi\left(\frac{3}{4}L - \log(8r)\right) + o(1), \qquad \mu_{a,\mathrm{all}} = \frac{3}{4}L - \log(8r) + o(1).$$

Consequently, for all sufficiently large N, uniformly in the parameters,

$$\frac{3L}{16} \le \mu_{a,\chi} \le L, \qquad \frac{5L}{8} \le \mu_{a,\mathrm{all}} \le L. \tag{22}$$

Thus either $Z_{a,\chi}(p) < L/8$ or $K_a(p) < L/2$ forces deviation from its own exact mean by at least $t_0 = L/16$.

3.3 A moment bound with explicit order dependence

Lemma 3.2. *Let B_i be finitely many independent Bernoulli variables of means w_i, and put $Y = \sum_i (B_i - w_i)$ and $\mu = \sum_i w_i$. For every integer $r \ge 1$,*

$$\mathbb{E}|Y|^{2r} \le 2(4\mu)^r r! + 2\,4^{2r}(2r)! \le 2(4\mu r)^r + 2(8r)^{2r}. \tag{23}$$

Proof. For $|s| \le 1$, independence and $\log(1+u) \le u$ yield

$$\log \mathbb{E}e^{sY} \le \mu(e^s - 1 - s) \le \mu s^2.$$

When $\mu = 0$ the claim is immediate. Otherwise use exponential Markov on each tail, choosing $|s| = \min(t/(2\mu), 1/2)$. This gives

$$\mathbb{P}(|Y| \ge t) \le 2\exp\left\{-\min\left(\frac{t^2}{4\mu}, \frac{t}{4}\right)\right\}.$$

Insert the bound into $2r\int_0^\infty t^{2r-1}\mathbb{P}(|Y| \ge t)\,dt$ and use $e^{-\min(u,v)} \le e^{-u} + e^{-v}$. The Gaussian and exponential integrals give respectively $2(4\mu)^r r!$ and $2\,4^{2r}(2r)!$. Finally $r! \le r^r$ and $(2r)! \le (2r)^{2r}$. □

For $\mu \le L$, $r \le cL$ and $t_0 = L/16$, this implies

$$\begin{aligned}\frac{\mathbb{E}|Y|^{2r}}{t_0^{2r}} &\le 2(1024r/L)^r + 2(128r/L)^{2r} \\ &\le 4\,2^{-4r} \le 4(\log N)^{-2c\log 2},\end{aligned} \tag{24}$$

where the last inequality uses $r \ge cL/2$, valid once $cL \ge 2$. Every constant in this estimate is independent of the growing order r.

3.4 The coefficient of a distinct support

Lemma 3.3. *Let $\mathcal{Q}$ be any finite set, $I_q \in \{0,1\}$, $T = \sum_{q\in\mathcal{Q}} I_q$, and let k be a nonnegative integer. Zero powers are interpreted as 1, including at zero, as polynomial identities; thus the case $k = 0$ is included. The multilinear expansion of $(T-\mu)^k$ is*

$$(T-\mu)^k = \sum_{\substack{S\subseteq\mathcal{Q}\\ |S|\le k}} \alpha_S \prod_{q\in S} I_q, \tag{25}$$

where, for $s = |S|$,

$$\alpha_S = \sum_{j=s}^{k} \binom{k}{j} (-\mu)^{k-j} s! \left\{ {j \atop s} \right\}. \tag{26}$$

For $\mu \ge 0$, $|\alpha_S| \le (\mu+s)^k$. The empty-support coefficient is $(-\mu)^k$.

Proof. Expand the power and use $I_q^j = I_q$ for $j \ge 1$. In a product of j indicators, using exactly the prescribed s indices is counted by the number of surjections onto those indices, namely $s!\left\{ {j \atop s} \right\}$. This proves (26). That number is at most s^j; applying the binomial theorem to the absolute values gives the asserted bound. For $s = 0$, only the term $j = 0$ remains, with the usual convention $\left\{ {0 \atop 0} \right\} = 1$. □

For our parameters $k = 2r$, $s \le 2r$, and $\mu \le L$, normalization before estimating gives the uniform bound

$$\frac{|\alpha_S|}{t_0^{2r}} \le [16(1+2c)]^{2r} \le 32^{2r} \le (\log N)^{\beta}, \qquad \beta = 10c\log 2 < 1. \tag{27}$$

It is this bound per distinct support, rather than a bound on the number of ordered tuples, that permits use of ordinary Bombieri–Vinogradov with r growing.

3.5 Transfer by distinct supports

Fix $a \in \mathcal{A}_N$ and a set $\mathcal{Q}$ of primes with $a < q \le Z$, all odd. Write $T(p) = \sum_{q\in\mathcal{Q}} I_q(p)$ and assume that its exact comparison mean is at most L. This includes the statistics in (20). In the comparison model, let B_q be independent with mean $1/(q-1)$, let $T^* = \sum B_q$, and let μ be the exact sum of their means. For each support S in Lemma 3.3, put $d(S) = \prod_{q\in S} q$. The simultaneous divisibility event for actual $p \in \mathcal{P}_N$ is exactly

$$p \equiv 1 \pmod 4, \qquad p \equiv -a \pmod{d(S)}.$$

Since the primes q exceed a, this is one reduced class modulo $4d(S)$. Write $P_0 = (\mathrm{Li}(2N) - \mathrm{Li}(N))/2$. Then

$$\sum_{p\in\mathcal{P}_N} \prod_{q\in S} I_q(p) = \frac{P_0}{\varphi(d(S))} + R_a(d(S)). \tag{28}$$

The model joint expectation is exactly $1/\varphi(d(S))$. For the empty support, $d(S) = 1$ and the remainder is $|\mathcal{P}_N| - P_0$; it is included in this formula.

Unique prime factorization ensures that distinct supports give distinct $d(S)$. Every such support has at most $2r$ primes, so

$$4d(S) \le 4Z^{2r} = 4N^{1/4}. \tag{29}$$

Using (17) with $A = 5$, the sum of absolute remainders over these supports is $O(N/(\log N)^5)$. In particular, it is not necessary to multiply this bound by the number of supports. Equations (25)–(28) and (27) yield

$$\frac{1}{t_0^{2r}} \sum_{p \in \mathcal{P}_N} (T(p) - \mu)^{2r} = P_0 \frac{\mathbb{E}(T^* - \mu)^{2r}}{t_0^{2r}} + O\big(N(\log N)^{-5+\beta}\big). \tag{30}$$

The signed expansion is used only to establish the equality and its error bound; the actual even moment on the left is nonnegative.

By Markov's inequality and (24), any event forcing $|T(p) - \mu| \geq t_0$ occurs for at most

$$4P_0\, 2^{-4r} + O\big(N(\log N)^{-5+\beta}\big) \tag{31}$$

primes. In particular, this applies to each failed threshold in (15), by (22).

Proposition 3.4 (Common exceptional-set estimate)**.** *For the parameters in* (14)*,*

$$|\mathcal{B}_N| \ll |\mathcal{P}_N| (\log N)^{-\delta}.$$

Proof. For each a we have $\varphi(a) - 1$ sector statistics and one total factor statistic. The sufficient-test bad set is $\mathcal{B}_N$ from (15). Its constituent events need not be disjoint; we only use a union bound. The number of statistics over all $a \in \mathcal{A}_N$ is at most

$$\sum_{a \leq 2(\log N)^\delta} \varphi(a) \ll (\log N)^{2\delta}. \tag{32}$$

A union bound in (31), using $P_0 \asymp |\mathcal{P}_N| \asymp N/\log N$, shows that the proportion of bad primes is

$$O\left((\log N)^{2\delta - 2c\log 2} + (\log N)^{-4+\beta+2\delta}\right) = O((\log N)^{-\delta}). \tag{33}$$

For the equality, note $c = 64\delta$, so $2c \log 2 > 3\delta$ and $4 - \beta > 3\delta$.

□

4 Proofs of the main results

Proof of Theorem 1.1. For $p \in \mathcal{P}_N \setminus \mathcal{B}_N$, every nonprincipal character modulo every $a \in \mathcal{A}_N$ has at least $L/8$ distinct prime divisors of x_a in its sector. Lemma 2.3 therefore gives, for every unit class,

$$\left| \frac{\varphi(a) C_{a,x_a}(b)}{\tau(x_a^2)} - 1 \right| \leq (\varphi(a) - 1) 3^{-L/8} \leq 2(\log N)^{\delta - \log 3/8} = 2(\log N)^{-\kappa}. \tag{34}$$

All additional prime factors and their powers are already included in C_{a,x_a}; no information about them beyond Lemma 2.3 is needed. Proposition 2.2, at $b = -p^{-1}$ and $b = -1$, now gives the two complete local counts. The first target depends on p, which is harmless because (34) holds for all unit classes for each individual good prime.

For sufficiently large N and $N < p \leq 2N$, $(\log p)^\delta \leq 2(\log N)^\delta$. The prescribed shifts $a_j = 3 + 4j$, $0 \leq j < J(p)$, are exactly those satisfying $3 \leq a_j \leq (\log p)^\delta$, so they belong to $\mathcal{A}_N$. Also $\log p \asymp \log N$, and hence the relative errors are $O((\log p)^{-\kappa})$, uniformly in j. They are eventually less than $1/2$, so the counts in the ratio are positive. Taking their ratio and their sum gives (5). Strict minimality distinguishes solutions at different positions.

Choose a fixed sufficiently large dyadic starting point and let $\mathcal{E}$ be the union of the sets $\mathcal{B}_N$ for disjoint dyadic intervals above it, together with the finite initial range of primes. This is

a fixed sufficient exceptional set. To estimate its counting function up to X, discard primes at most $\sqrt{X}$; their number is negligible compared with $X/(\log X)^{1+\delta}$. For the remaining dyadic intervals $N \gg \sqrt{X}$, the logs are uniformly comparable to $\log X$. Proposition 3.4 and the prime number theorem give

$$\sum_N |\mathcal{B}_N| \ll (\log X)^{-\delta} \sum_N |\mathcal{P}_N| \ll X/(\log X)^{1+\delta}.$$

A possible final partial interval is bounded by its full dyadic interval. This proves (2). □

4.1 A broader total-factor window

Keep the sector window unchanged, and also put

$$W = 2(\log N)^\delta, \qquad \widetilde{K}_a(p) = \sum_{W<q\le Z} I_q(p), \qquad \widetilde{\mu} = \sum_{W<q\le Z} \frac{1}{q-1}. \tag{35}$$

Every prime in this wider window exceeds every $a \in \mathcal{A}_N$. Its exact comparison mean does not depend on a. The case $a = 1$ of (16), followed by partial summation, gives

$$\sum_{W<q\le Z} \frac{1}{q-1} = \int_W^Z \frac{dt}{t\log t} + O((\log W)^{-2} + W^{-1}).$$

For example, $H = 3$ suffices for the integrated prime-counting error; replacing $1/q$ by $1/(q-1)$ costs $O(W^{-1})$. Since $W \to \infty$, this proves

$$\widetilde{\mu} = L - \log(8r) - \log(\log 2 + \delta L) + o(1) = L - O(\log L). \tag{36}$$

In particular, $\widetilde{\mu} \le L$ eventually. For every fixed $0 < \theta < 15/16$ it is eventually at least $(\theta + 1/16)L$. Thus $\widetilde{K}_a(p) < \theta L$ forces a deviation of at least $t_0 = L/16$ from its own exact mean. The lower endpoint requires only the ordinary prime number theorem.

Proposition 4.1 (An enlarged sufficient-test bad set)**.** *Fix $0 < \theta < 15/16$ and define*

$$\mathcal{B}_N^* = \mathcal{B}_N \cup \bigcup_{a\in\mathcal{A}_N} \{p \in \mathcal{P}_N : \widetilde{K}_a(p) < \theta L\}. \tag{37}$$

Then, for sufficiently large N depending on θ,

$$|\mathcal{B}_N^*| \ll_\theta |\mathcal{P}_N|(\log N)^{-\delta}.$$

Proof. Equation (36) gives $\widetilde{\mu} \le L$ and the required deviation $L/16$. Thus the normalized model moment is at most $4\,2^{-4r}$ by (24). Every distinct support still has normalized coefficient at most $(\log N)^\beta$ by (27), and its modulus is at most $4Z^{2r} = 4N^{1/4}$ by (29). The lower endpoint has changed, but all support primes exceed a and are odd. Consequently the same reduced-class count and the same unweighted BV remainder sum apply. Formula (31) bounds the number of added failures for each a.

There are $|\mathcal{A}_N| \ll (\log N)^\delta$ new statistics, compared with the $O((\log N)^{2\delta})$ original statistics. The enlarged union therefore still has bad proportion

$$O\Big((\log N)^{2\delta - 2c\log 2} + (\log N)^{-4+\beta+2\delta}\Big) = O((\log N)^{-\delta}).$$

Here $c = 64\delta$, $2c\log 2 > 3\delta$, and $4 - \beta > 3\delta$. The necessary onset, and the constant after absorbing an initial range, may depend on the fixed θ. □

Proof of Corollary 1.2. On $\mathcal{P}_N \setminus \mathcal{B}_N^*$ there are at least θL distinct prime factors of x_a in the wider window. Each contributes at least 3 to $\tau(x_a^2)$, so

$$\tau(x_a^2) \geq 3^{\widetilde{K}_a(p)} \geq (\log N)^{\theta \log 3}.$$

The relative error in (34) is eventually at most $1/2$. The exact counts then give dyadic lower bounds with denominators $2\varphi(a)$ for Type I and $4\varphi(a)$ for Type II. For fixed θ, converting $\log N$ to $\log p$ costs at most another factor two for sufficiently large N. This proves (6) on the enlarged good set. Define $\mathcal{E}_\theta$ from the dyadic $\mathcal{B}_N^*$ and a sufficiently large finite initial range, including $\mathcal{E}$. The same dyadic summation as above proves the claimed counting bound. The original asymptotics remain valid because the enlarged bad set contains the original one. □

Proof of Corollary 1.3. For fixed J_0, the prescribed positions are included once

$$p \geq P_{J_0} := \exp((4J_0 - 1)^{1/\delta}),$$

since this is exactly the condition $J(p) \geq J_0$. Theorem 1.1, respectively Corollary 1.2, applies above this threshold. The finitely many smaller primes are absorbed in the constants, which are allowed to depend on J_0 and, for the multiplicity assertion, on θ. No effective starting point is obtained by this deduction. □

5 Arithmetic interpretation and parameters

5.1 Quadratic reciprocity

The following observation interprets one of the sector tests already used. Here $p \equiv 1 \pmod 4$ is prime, $a \equiv 3 \pmod 4$, $3 \leq a < p$, $x = (p + a)/4$, and $x = \prod q^{e_q}$, $d = \prod q^{j_q}$ in the notation of Lemma 2.1.

Proposition 5.1 (Reciprocity interpretation)**.** *For every prime factor $q \mid x$,*

$$\left(\frac{q}{p}\right) = \left(\frac{q}{a}\right), \tag{38}$$

where the right hand side is the Jacobi symbol. The character $\chi_a(n) = (n/a)$ on the unit group modulo a is nonprincipal and odd. Thus the corresponding sector count is exactly

$$Z_{a,\chi_a}(p) = \#\{Q < q \leq Z : q \mid x,\ q \text{ prime},\ (q/p) = -1\}.$$

For each Type II witness $d \mid x^2$, the primitive resource is

$$R = uv = \frac{dx}{(d, x)^2} = \prod_{q \mid x} q^{|j_q - e_q|} \mid x.$$

It satisfies $(R/p) = -1$, and in particular $R \geq n_-(p)$, the least positive quadratic nonresidue modulo p.

Proof. For odd $q \mid x$, $p \equiv -a \pmod q$ and quadratic reciprocity give

$$(q/p) = (p/q) = (-a/q) = (-1/q)(a/q) = (q/a),$$

because $p \equiv 1 \pmod 4$ and $a \equiv 3 \pmod 4$ make the two signs in the last step cancel. If $q = 2$, then $p \equiv -a \pmod 8$, so the supplementary law gives $(2/p) = (2/a)$. The value $\chi_a(-1) = -1$ proves that χ_a is odd and nonprincipal. Its values on primes dividing x are ± 1, so its sector is exactly the negative-value set just displayed.

The formula for R follows prime by prime from Lemma 2.1. In Type II, $u/v \equiv -1 \pmod{a}$. As Jacobi values on units are ± 1,

$$\chi_a(uv) = \chi_a(u)\chi_a(v)^{-1} = \chi_a(uv^{-1}) = \chi_a(-1) = -1.$$

Multiplicativity and (38) now give $(R/p) = -1$. The least-nonresidue bound follows from its definition. □

This identifies the quadratic-character sector condition with prime-specific arithmetic: on the sufficient good set, every prescribed x_a has at least $L/8$ distinct nonresidue prime factors in the specified window. The resource uv is a divisor of x_a attached to the primitive ratio of a Type II witness.

5.2 Parameter budget

The catalogue and exceptional-set exponents can be separated. Keep $c = 2^{-14}$, $r = \lfloor cL \rfloor$, and $\beta = 10c \log 2$. For fixed $\eta, \sigma > 0$, a sufficient budget for a catalogue $a \le 2(\log N)^\eta$ and an exceptional-proportion saving $(\log N)^{-\sigma}$ is

$$\begin{aligned} &\eta < \min\{1/4,\ (\log 2)/16,\ (\log 3)/8\}, \\ &2\eta + \sigma < 2c \log 2, \qquad 2\eta + \sigma < 4 - \beta. \end{aligned} \tag{39}$$

The first line makes the errors in the uniform mean estimates and both auxiliary Fourier estimates tend to zero. The full-box estimate then has relative error $O((\log N)^{\eta - \log 3/8})$. There are $O((\log N)^{2\eta})$ original statistics; the second line pays for their union in the moment and BV estimates. The broader window $W = 2(\log N)^\eta$ adds only $O((\log N)^\eta)$ statistics, and its mean is still $L - O_\eta(\log L)$. Its threshold argument works for every fixed $\theta < 15/16$. Constants and onsets may depend on these fixed parameters. The present choice $\eta = \sigma = \delta$ satisfies these inequalities with room to spare. This is a sufficient budget, not an optimization.

The all-class estimate supplies both targets -1 and $-p^{-1}$ without another union over residue classes. The resulting $2:1$ ratio holds on the common good set. It can fail at an individual position: $(p, a, x) = (53, 7, 15)$ has $F_{\mathrm{I}} = 2$ and $F_{\mathrm{II}} = 0$.

The supplementary exact checks cover the divisor dictionary, complete local counts, prime-power boxes, recovery multiplicities, and the coefficient budgets. Their finite ranges and arithmetic conventions are recorded with the verification code.

A Squarefree subfamilies and one-sided recovery

The complete-count argument also supplies useful squarefree subfamilies. Three-way allocation of coprime factors, and its 3^K count, already occurs in Browning and Elsholtz [1, Section 2] and Elsholtz and Planitzer [2, Section 7].

A.1 Binary and ternary character estimates

Lemma A.1 (Binary subset products)**.** *Let G be a finite abelian group, $K \in \mathbb{Z}_{\ge 0}$, $m \ge 0$, and $g_1, \dots, g_K \in G$. Suppose that every nonprincipal character has at least m indices i with $\Re\chi(g_i) \le 0$. If $D(b)$ counts subsets with product b, then*

$$\left| \frac{|G| D(b)}{2^K} - 1 \right| \le (|G| - 1) 2^{-m/2} \qquad (b \in G). \tag{40}$$

If the right-hand side is at most $1/2$, then $D(b) \ge 2^{K-1}/|G|$ for every $b \in G$.

Proof. Let $\nu(b) = 2^{-K} D(b)$, and use the Fourier convention

$$\widehat{\nu}(\chi) = \sum_{b \in G} \nu(b)\chi(b), \qquad \nu(b) = \frac{1}{|G|} \sum_{\chi \in \widehat{G}} \widehat{\nu}(\chi)\overline{\chi(b)}.$$

Expanding over the choice of each index gives the exact identity

$$\widehat{\nu}(\chi) = \prod_{i=1}^{K} \frac{1 + \chi(g_i)}{2}.$$

Every factor has absolute value at most 1. On the specified m indices,

$$\left|\frac{1 + \chi(g_i)}{2}\right|^2 = \frac{1 + \operatorname{Re}\chi(g_i)}{2} \le \frac{1}{2}.$$

The nonprincipal Fourier coefficients therefore have absolute value at most $2^{-m/2}$. The principal coefficient is exactly 1, so Fourier inversion and the triangle inequality give (40). For the trivial group the nonprincipal sum is empty and the error is zero. When $K = 0$, the only subset is the empty subset; the same formulas apply. □

Lemma A.2. *Let G be a finite abelian group, $K \in \mathbb{Z}_{\ge 0}$, $m \ge 0$, and $g_1, \dots, g_K \in G$. For $b \in G$, let $T(b)$ count assignments $\epsilon_i \in \{-1, 0, 1\}$ with $\prod_i g_i^{\epsilon_i} = b$. If every nonprincipal character has at least m indices i with $\operatorname{Re}\chi(g_i) \le 0$, then*

$$\left|\frac{|G|T(b)}{3^K} - 1\right| \le (|G| - 1)3^{-m}. \tag{41}$$

In particular, if the right-hand side is at most $1/2$, then $T(b) \ge 3^K/(2|G|)$ for every $b \in G$.

Proof. The normalized character transform of the three choices at index i is

$$\frac{1 + \chi(g_i) + \chi(g_i)^{-1}}{3} = \frac{1 + 2\operatorname{Re}\chi(g_i)}{3}.$$

It has absolute value at most 1 in general and at most $1/3$ when $\operatorname{Re}\chi(g_i) \le 0$. Hence every nonprincipal coefficient has absolute value at most 3^{-m}, while the principal coefficient is 1. Fourier inversion proves (41); the trivial-group and $K = 0$ cases are immediate. □

A.2 Simultaneous window supply

For a unit class $b \pmod a$, let $D_{a,p}(b)$ count subsets of $\mathcal{Q}_a(p)$ whose product is b modulo a, with the empty product equal to 1. Let $T_{a,p}(b)$ count ordered pairs (u, v) obtained as products of disjoint subsets of $\mathcal{Q}_a(p)$ with $u/v \equiv b \pmod a$. The ratio is taken in the unit group. There are three choices per window prime, and every such pair satisfies $(u, v) = 1$ and $uv \mid x_a$.

Theorem A.3 (Auxiliary squarefree-window supply)**.** *With (14) and (15),*

$$|\mathcal{B}_N| \ll |\mathcal{P}_N|(\log N)^{-\delta}.$$

Every $p \in \mathcal{P}_N \setminus \mathcal{B}_N$ satisfies, simultaneously for all $a \in \mathcal{A}_N$ and all unit classes $b \pmod a$,

$$\begin{aligned} K_a(p) \ge L/2, \qquad D_{a,p}(b) &\ge \frac{2^{K_a(p)-1}}{\varphi(a)}, \\ T_{a,p}(b) \ge \frac{3^{K_a(p)}}{2\varphi(a)} \ge \frac{(\log N)^{\gamma_0}}{2\varphi(a)}, \qquad \gamma_0 &= \frac{\log 3}{2}. \end{aligned} \tag{42}$$

Proof. The exceptional-set estimate is Proposition 3.4. For every remaining p, all $a \in \mathcal{A}_N$ simultaneously satisfy

$$K_a(p) \geq L/2, \qquad Z_{a,\chi}(p) \geq L/8 \quad (\chi \neq \chi_0).$$

Apply Lemma A.1 and Lemma A.2 to the residues of the distinct primes in $\mathcal{Q}_a(p)$. The binary relative error is at most

$$(\varphi(a)-1)2^{-L/16} \leq 2(\log N)^{\delta-\log 2/16} = o(1),$$

while the ternary relative error is at most

$$(\varphi(a)-1)3^{-L/8} \leq 2(\log N)^{\delta-\log 3/8} = o(1).$$

Both are eventually at most $1/2$, uniformly in a. Thus every unit class satisfies both the displayed binary bound and $T_{a,p}(b) \geq 3^{K_a(p)}/(2\varphi(a))$. Since $3^{L/2} = (\log N)^{\gamma_0}$, the last ternary lower bound in (42) follows. The selected primes are distinct actual divisors of x_a; this verifies the integer-divisor interpretation as well as both residue-class counts. □

The exponent $\gamma_0 = (\log 3)/2$ concerns this selected window; Corollary 1.2 uses the broader total-factor window.

A.3 A one-sided Type II construction

Lemma A.4. *Let p be prime, $a \equiv 3 \pmod 4$, $3 \leq a < p$, and $p \equiv 1 \pmod 4$. Put $x = (p+a)/4$. If d is a positive divisor of x and $d \equiv -1 \pmod a$, then, with*

$$h = \frac{d+1}{a}, \qquad y = \frac{phx}{d}, \qquad z = phx, \tag{43}$$

we have a Type II solution $x < y < z$ of (1). Distinct such d give distinct unordered triples at the fixed position x.

Proof. All quantities in (43) are positive integers. Since $d \equiv -1 \pmod a$ and $a \geq 3$, we have $d \geq a-1 \geq 2$. Moreover

$$\frac{1}{x}+\frac{1}{y}+\frac{1}{z} = \frac{1}{x}+\frac{d+1}{phx} = \frac{p+a}{px} = \frac{4}{p}.$$

Both y and z are multiples of p and at least p, whereas $x < p/2$. Hence x is strictly smallest, $p \nmid x$, and $z/y = d > 1$. The ratio of the larger to the smaller of the remaining two denominators recovers d from the unordered triple, proving injectivity. □

B Fixed catalogues from earlier inputs

The following consequence explains the distinction between the fixed-catalogue precedent and the uniform quantitative application.

Proposition B.1 (Fixed-catalogue consequence)**.** *Let $\mathcal{A} \subset \{3, 7, 11, \ldots\}$ be fixed and finite. On a relative-density-one set of primes $p \equiv 1 \pmod 4$, simultaneously for $a \in \mathcal{A}$ and $x_a = (p+a)/4$,*

$$F_{\mathrm{I}}(p, x_a) \sim \frac{\tau(x_a^2)}{\varphi(a)}, \qquad F_{\mathrm{II}}(p, x_a) \sim \frac{\tau(x_a^2)}{2\varphi(a)}.$$

For every fixed $0 < \varepsilon < \log 3$, each count is also $\gg_{\mathcal{A},\varepsilon} (\log p)^{\log 3-\varepsilon}$ on a possibly smaller relative-density-one set.

Proof. For fixed a and a nonprincipal character $\chi \bmod a$, define the nonnegative strongly additive function

$$w_{a,\chi}(n) = \sum_{\substack{q \mid n \\ q \nmid a}} \mathbf{1}_{\Re\chi(q) \le 0}.$$

Let $\sigma_\chi = |S_\chi|/\varphi(a)$. Lemma 3.1 gives $\sigma_\chi \ge 1/3$. The prime number theorem in fixed arithmetic progressions gives

$$\sum_{q \le N} \frac{w_{a,\chi}(q)}{q-1} = \sigma_\chi \log\log N + O_a(1), \qquad \max_{q \le N} w_{a,\chi}(q) \le 1.$$

The fixed-shift normal-order theorem of Barban and Elliott, recalled as Lemma A.3 of [6], therefore applies to $p+a$. After a finite union over (a,χ), its conclusion holds for all $w_{a,\chi}(p+a)$ outside $o(\pi(X))$ primes up to X. Since $\pi(X;4,1) \sim \pi(X)/2$, restricting to $p \equiv 1 \pmod 4$ preserves relative density one. On this progression x_a is an integer, and dividing $p+a$ by 4 removes at most one distinct prime factor. Consequently, on a relative-density-one set in this progression,

$$w_{a,\chi}(x_a) \ge \tfrac{1}{4}\log\log p \qquad (a \in \mathcal{A},\ \chi \ne \chi_0).$$

Lemma 2.3 now yields, uniformly over this fixed catalogue and all unit classes b,

$$C_{a,x_a}(b) = \frac{\tau(x_a^2)}{\varphi(a)}\left(1 + O_{\mathcal{A}}((\log p)^{-(\log 3)/4})\right).$$

Proposition 2.2 gives the asserted complete counts.

Apply the same fixed-shift normal-order theorem to $w = \omega$. Restricting to the same progression and dividing by 4 as above gives, for all $a \in \mathcal{A}$ simultaneously, $\omega(x_a) = (1 + o(1))\log\log p$ on a relative-density-one set there. Since $\tau(x_a^2) \ge 3^{\omega(x_a)}$, the claimed lower bounds follow, with any fixed positive ε providing the exponent margin. □

This deduction uses finitely many fixed shifts and supplies an $o(\pi(X))$ exceptional set. The common logarithmic-power estimate for a specified growing catalogue requires the uniform calculation in the main text or Appendix C.

B.1 The per-class divisor route

For fixed J_0 and $0 < \varepsilon < \log 2$, the per-class divisor estimates in the final part of the proof of Bettin et al. [7, Lemma 4.9 and equation (4.3)] also imply

$$F_{\mathrm{II}}(p, (p+3)/4 + j) \gg_{J_0,\varepsilon} (\log p)^{\log 2 - \varepsilon} \qquad (0 \le j < J_0) \tag{44}$$

for a density-one set of the relevant primes. This is an extraction from their proof, not an Erdős–Straus theorem named there. Indeed, for a fixed $a = 3 + 4j$, take their shift $p_0 = -a$, modulus $4a$, and unit class $2a - 1$. Their estimate supplies $\gg_{a,\varepsilon} (\log p)^{\log 2 - \varepsilon}$ divisors $d \mid p + a$ in that class. Such d are odd, hence divide $x = (p+a)/4$, and satisfy $d \equiv -1 \pmod a$. Lemma A.4 converts different d into different solutions. A finite union over the fixed shifts proves (44). This argument does not extract uniform constants for growing J_0 or the explicit exceptional-set estimate (2).

C An alternative transfer by Elliott's model

Elliott's Bernoulli approximation, in the explicit form of [10, Theorem 1.1], can replace the moment transfer. We verify its hypotheses uniformly for the growing shifts. In the form we use, let V take values in $[1,T]\cap\mathbb{N}$, let $2\le y\le T$, let $\mathcal{Q}$ be a set of primes at most y, and let $g:\mathbb{N}\to[0,1]$ be multiplicative, vanish off $\langle\mathcal{Q}\rangle$, and satisfy $g(q)<1$ for $q\in\mathcal{Q}$. Let B_q be independent Bernoulli variables with means $g(q)$ and put $S=\sum_{q\le y}g(q)\log q/(1-g(q))$. For $S>0$ and $\log z\ge 8\max\{\log y,S\}$, Elliott's bound is

$$\begin{aligned} &d_{\mathrm{TV}}\big((\mathbf{1}_{q\mid V})_{q\in\mathcal{Q}},(B_q)_{q\in\mathcal{Q}}\big)\\ &\quad\le 10\exp\left(-\frac{\log z}{8\log y}\log\frac{\log z}{S}\right)+12\sum_{\substack{d\le z^4\\ d\in\langle\mathcal{Q}\rangle}}\mu^2(d)4^{\omega(d)}\big|\mathbb{P}(d\mid V)-g(d)\big|. \end{aligned}\tag{45}$$

Keep (14) and (35), and put

$$\begin{gathered} y=Z,\qquad z=N^{1/16},\qquad H=|\mathcal{P}_N|,\\ P_0=\frac{\mathrm{Li}(2N)-\mathrm{Li}(N)}{2},\qquad \mathcal{Q}_N=\{q\text{ prime}:W<q\le Z\}. \end{gathered}$$

For $a\in\mathcal{A}_N$, choose p uniformly from $\mathcal{P}_N$; then $x_a=(p+a)/4$ is an integer in $[1,N]$ for sufficiently large N. Let $(B_q)_{q\in\mathcal{Q}_N}$ be independent Bernoulli variables with $\mathbb{P}(B_q=1)=1/(q-1)$.

Proposition C.1 (Uniform Bernoulli approximation)**.** *For every fixed $B>0$,*

$$\sup_{a\in\mathcal{A}_N} d_{\mathrm{TV}}\Big((\mathbf{1}_{q\mid x_a})_{q\in\mathcal{Q}_N},(B_q)_{q\in\mathcal{Q}_N}\Big)\ll_B(\log N)^{-B}.\tag{46}$$

The implied constant is independent of the growing shift a.

Proof. Define the multiplicative function g by $g(1)=1$, $g(q^j)=1/\varphi(q^j)$ for $q\in\mathcal{Q}_N$, $j\ge1$, and $g(d)=0$ outside the semigroup $\langle\mathcal{Q}_N\rangle$. This takes values in $[0,1]$ and has $g(q)<1$ at every window prime. The parameter in Elliott's theorem satisfies

$$S=\sum_{W<q\le y}\frac{\log q}{q-2}\le\log y+O(1)\le 2\log y.$$

Here Mertens' estimate for $\sum_{q\le y}(\log q)/q$ suffices; the correction from q to $q-2$ has a bounded sum over $q\ge3$. Since $\log z/\log y=r/2$, the hypothesis $\log z\ge8\max\{\log y,S\}$ holds once $r\ge32$. The first term of Elliott's bound is then at most

$$10\exp\left(-\frac{r}{16}\log\frac{r}{4}\right)\ll_B(\log N)^{-B}.\tag{47}$$

The last assertion follows from $r\sim c\log\log N$.

It remains to bound the theorem's weighted remainder sum, whose range is $d\le z^4=D=N^{1/4}$ and whose weight is $\mu^2(d)4^{\omega(d)}$. Every $d\in\langle\mathcal{Q}_N\rangle$ is odd and coprime to every $a\in\mathcal{A}_N$. The congruences $p\equiv1\pmod4$ and $p\equiv-a\pmod d$ specify one reduced class modulo $4d$. Consequently, on writing

$$\#\{p\in\mathcal{P}_N:d\mid x_a\}=\frac{P_0}{\varphi(d)}+R_a(d),$$

we have the exact normalization

$$\mathbb{P}(d \mid x_a) - g(d) = \frac{R_a(d)}{H} + \frac{P_0 - H}{H\varphi(d)}. \tag{48}$$

Define

$$E_d = \max_{(b,4d)=1} \left| \pi(2N; 4d, b) - \pi(N; 4d, b) - \frac{\mathrm{Li}(2N) - \mathrm{Li}(N)}{\varphi(4d)} \right|.$$

Then $|R_a(d)| \le E_d$ uniformly in a. Equation (17) gives, for every fixed $A > 0$

$$\sum_{d \le D} E_d \ll_A N(\log N)^{-A}. \tag{49}$$

Indeed, the moduli $4d \le 4N^{1/4}$ are strictly within its range. The trivial count of integers in a residue class also gives $E_d \ll N/\varphi(d)$ for $d \le D$. For each fixed positive integer j, an Euler product and Mertens' estimate yield

$$\sum_{d \le D} \frac{\mu^2(d) j^{\omega(d)}}{\varphi(d)} \le \prod_{q \le D} \left(1 + \frac{j}{q-1}\right) \ll_j (\log(2D))^j. \tag{50}$$

Cauchy–Schwarz, followed by (49) and (50) with $j = 16$, therefore gives

$$\begin{aligned} \sum_{\substack{d \le D \\ d \in \langle \mathcal{Q}_N \rangle}} \mu^2(d) 4^{\omega(d)} E_d &\le \left(\sum_{d \le D} \mu^2(d) 16^{\omega(d)} E_d \right)^{1/2} \left(\sum_{d \le D} E_d \right)^{1/2} \\ &\ll_A N(\log N)^{8 - A/2}. \end{aligned} \tag{51}$$

Since $H \asymp N/\log N$, its normalized contribution is $O_A((\log N)^{9-A/2})$. Also $|P_0 - H| \le E_1 \ll_A N(\log N)^{-A}$. Its contribution in (48), using (50) with $j = 4$, is $O_A((\log N)^{5-A})$. Taking, for example, $A = 2B + 20$ bounds both remainder contributions by $O_B((\log N)^{-B})$. Together with (47) and Elliott's absolute factor 12, this proves (46). □

C.1 The common good set

The sector and total-factor statistics in (20) and (35) are functions of this one divisor vector. Their model means satisfy (22) and (36); every failed threshold forces a deviation of at least $t_0 = L/16$. Lemma 3.2 and (24) bound each model failure probability by $4\,2^{-4r}$.

For each fixed a, take the union of all its failed sector tests and its two failed total tests as a single event of the vector in (46). The comparison error is paid once for this union, not once per character. Hence the enlarged bad set $\mathcal{B}_N^*$ satisfies

$$\begin{aligned} \frac{|\mathcal{B}_N^*|}{H} &\le \sum_{a \in \mathcal{A}_N} \left(4(\varphi(a) + 1) 2^{-4r} + O_B((\log N)^{-B}) \right) \\ &\ll_{B,\theta} (\log N)^{2\delta - 2c\log 2} + (\log N)^{\delta - B} \ll_\theta (\log N)^{-\delta}, \end{aligned} \tag{52}$$

where $B > 2\delta$ is fixed and $2c\log 2 > 3\delta$. Omitting the wider total test gives Proposition 3.4. Including it gives Proposition 4.1, and the deterministic argument of Section 4 completes the same local-count and multiplicity conclusions. For each a this is a joint approximation of its divisor vector; shifts are combined by a union bound.

D The distribution inputs in the required normalization

We spell out the standard consequences used in Section 3. Write Λ_{vM} for the von Mangoldt function, and let

$$\psi(t;\chi)=\sum_{n\le t}\Lambda_{\mathrm{vM}}(n)\chi(n),\qquad \vartheta(t;a,b)=\sum_{\substack{q\le t\\ q\equiv b\ (a)}}\log q.$$

For a reduced class, $\psi(t;a,b)$ denotes the analogous sum of $\Lambda_{\mathrm{vM}}(n)$ over all integers in that class.

D.1 Siegel–Walfisz and the harmonic means

The character version on p. 2 of [13] gives, for fixed $B>0$ and every character modulo $a\le(\log t)^B$,

$$\psi(t;\chi)-\mathbf{1}_{\chi=\chi_0}t\ll_B t\exp(-c_B\sqrt{\log t})$$

with some $c_B>0$. Character orthogonality, equation (6) there, gives the same error for a reduced progression: the number of characters cancels the factor $1/\varphi(a)$.[1]

The prime powers contribute $O(\sqrt{t}\log^2 t)$, uniformly in the progression. Partial summation (equation (4), p. 1 of [13]) then gives (16). For completeness, when $a\le\log t$, split the integral at $\sqrt{t}$. In the upper part, $a\le(\log u)^2$ for all $u\ge\sqrt{t}$ once t is large, so the character bound applies with the single fixed choice $B=2$. In the lower part the trivial bound contributes at most $O(\sqrt{t}\log^2 t)$. Exponential decay absorbs every prescribed fixed power $(\log t)^{-H}$. The usual possible ineffectivity of Siegel–Walfisz remains; it does not create an unproved input or an effective onset.

In our application $a\le 2(\log N)^\delta\le\log Q$ eventually. Applying partial summation to $1/q$ on $(Q,Z]$ with $H=9$ bounds the integrated error by $O((\log Q)^{-8})$. Replacing $1/q$ by $1/(q-1)$ costs at most $O(\sum_{n>Q}n^{-2})=O(Q^{-1})$. This proves (19) with constants independent of a, its unit class, and the growing order r.

D.2 The dyadic prime-counting form of Bombieri–Vinogradov

The two-term bound on p. 3 of [13] states, for fixed $A_0>0$,

$$\sum_{m\le M}\max_{(v,m)=1}\sup_{1\le u\le T}\left|\psi(u;m,v)-\frac{u}{\varphi(m)}\right|\ll_{A_0}T(\log T)^{-A_0}+T^{1/2}M\big(\log(TM)\big)^6.$$

Use $T=2N$, $M=4N^{1/4}$, and denote the maximum and supremum for a fixed m by E_m. The second term is $O(N^{3/4}(\log N)^6)$ and is smaller than every fixed logarithmic saving relative to N.

For $N\le u\le 2N$, put $\Delta_m(u;v)=\vartheta(u;m,v)-u/\varphi(m)$. Removing prime powers gives

$$\max_{(v,m)=1}\sup_{N\le u\le 2N}|\Delta_m(u;v)|\le E_m+O(\sqrt{N}\log^2 N).$$

The exact dyadic partial-summation identity is

$$\pi(2N;m,v)-\pi(N;m,v)-\frac{\mathrm{Li}(2N)-\mathrm{Li}(N)}{\varphi(m)}=\frac{\Delta_m(2N;v)}{\log(2N)}-\frac{\Delta_m(N;v)}{\log N}+\int_N^{2N}\frac{\Delta_m(u;v)}{u\log^2 u}\,du.$$

[1] We use the character statement; the preceding progression display in that version of the notes omits the factor t in its error term.

After maximizing and summing, its absolute value is bounded by

$$\ll \frac{1}{\log N}\left(\sum_{m\le M} E_m + M\sqrt{N}\log^2 N\right).$$

Choose $A_0 = A+1$. Both terms of size $N^{3/4}$ are absorbed, giving (17). The proof uses only the interval $[N, 2N]$ at the partial-summation step, so no modulus-level assumption at much smaller integration variables is needed.

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